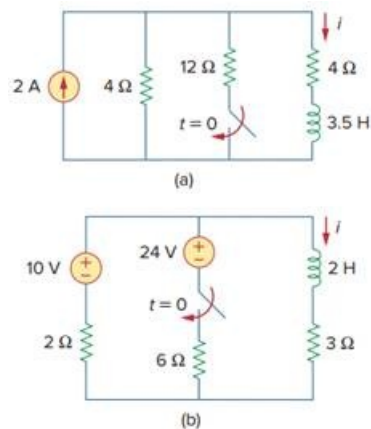


Problem

Obtain the inductor current for both $t < 0$ and $t > 0$ in each of the circuits in Fig. 7.120.

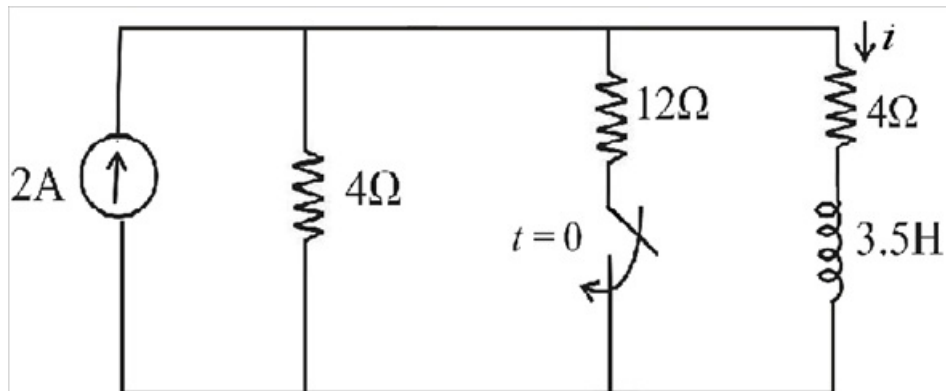
Fig. 7.120:



Step-by-step solution

Step 1 of 12

(a)



Step 2 of 12

At $t \rightarrow 0^-$ $i(0^-)$ is given by

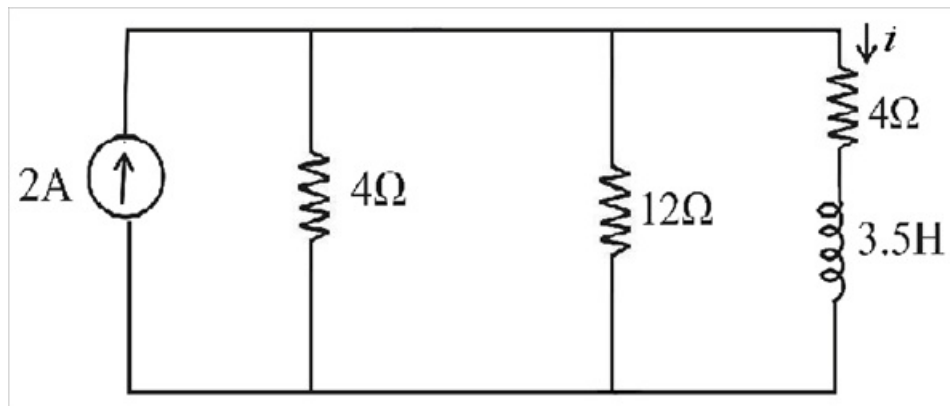
$$i(0^-) = 2 \times \frac{4}{8}$$

$$\therefore i(0^-) = 1 \text{ A}$$

$$\therefore \boxed{i(t) = 1 \text{ A}} \text{ for } t < 0$$

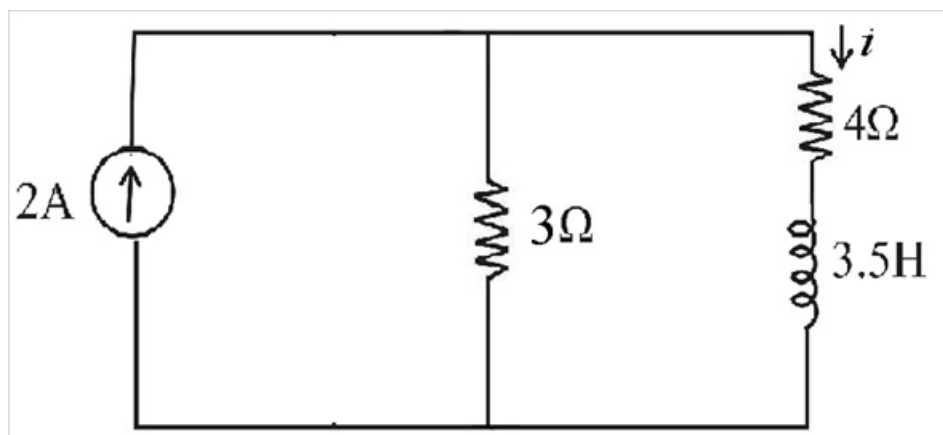
Step 3 of 12

Now at $t \rightarrow 0^+$ circuit is



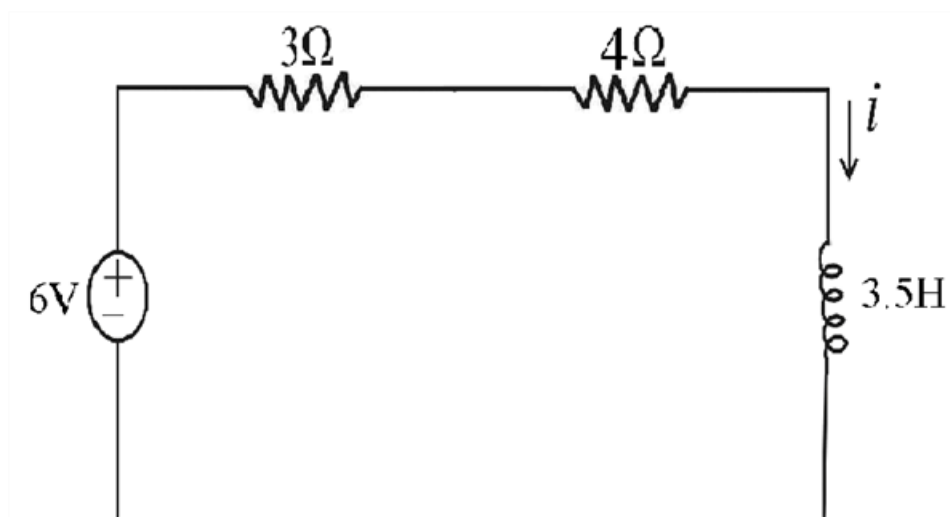
Step 4 of 12

The equivalent circuit is,



Step 5 of 12

Converting the circuit by source transformation we get,



Step 6 of 12

The response of the system is given by

$$i(t) = i(\infty) - [i(\infty) - i(0)]e^{-\frac{t}{\tau}}$$

$$i(\infty) = \frac{6}{3+4}$$

$$\therefore i(\infty) = 0.857 \text{ A}$$

$$\tau = \frac{L}{R_{eq}}$$

$$\tau = \frac{3.5}{3+4}$$

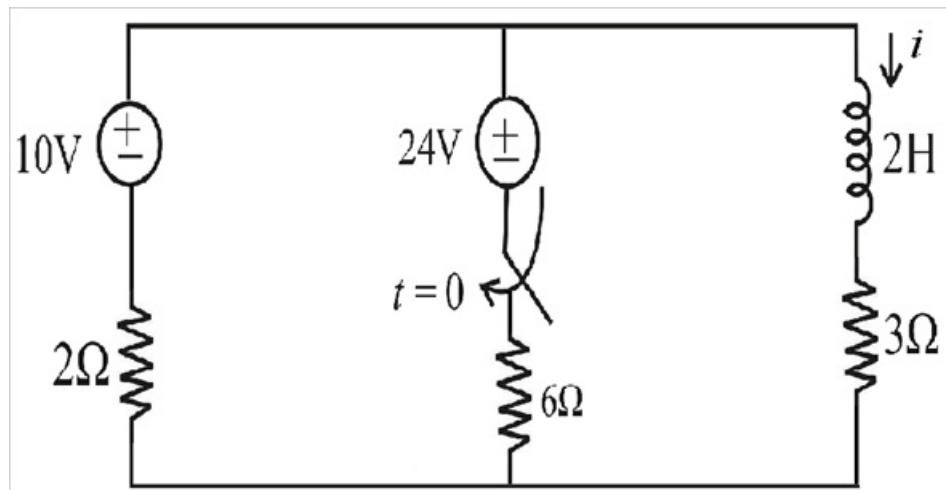
$$\therefore \tau = 0.5 \text{ s}$$

$$i(t) = 0.857 - [0.857 - 1]e^{-\frac{t}{0.5}}$$

$$\therefore \boxed{i(t) = 0.857 + 0.143 e^{-2t} \text{ A}} \text{ for } t > 0$$

Step 7 of 12

(b)



Step 8 of 12

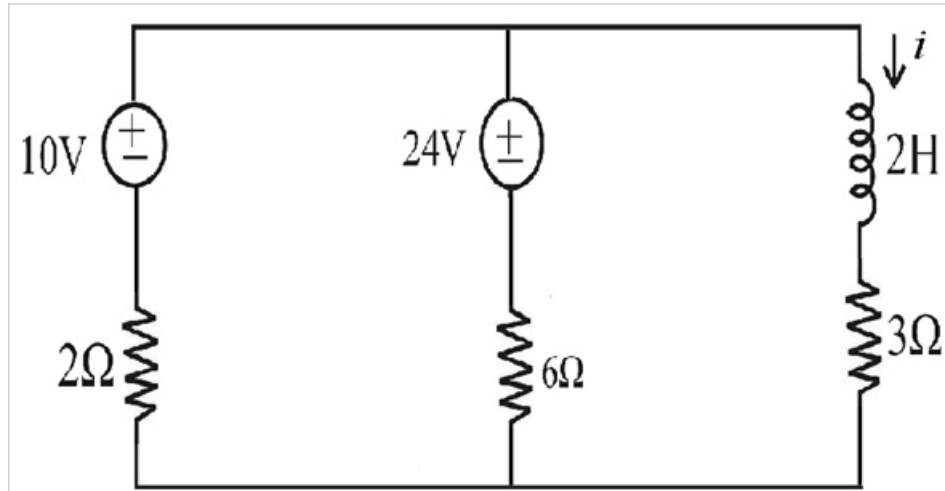
At $t = 0^-$ the current through inductor is,

$$i(0^-) = \frac{10}{2+3}$$

$$\therefore i(0^-) = 2 \text{ A}$$

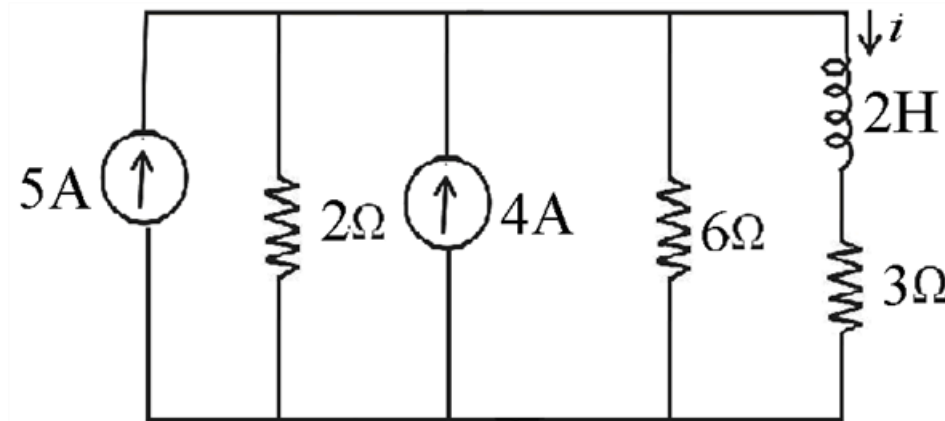
$$\therefore \boxed{i(t) = 2 \text{ A}} \text{ for } t < 0$$

After switch is closed,



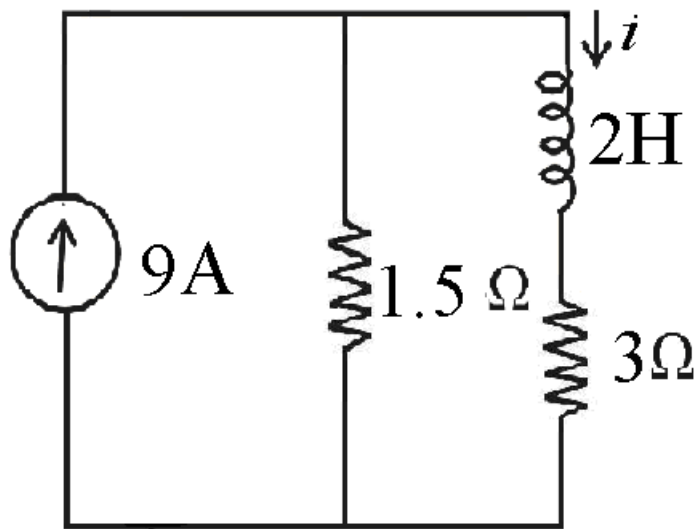
Step 9 of 12

By source transformation the circuit reduces to



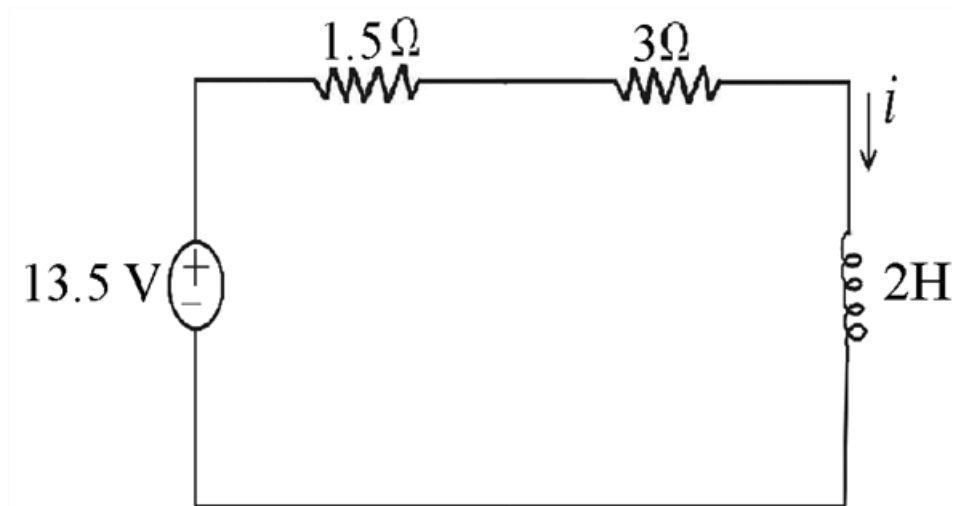
Step 10 of 12

The equivalent circuit is



Step 11 of 12

Converting the circuit by source transformation we get,



Step 12 of 12

The response of the system is given by

$$i(t) = i(\infty) - [i(\infty) - i(0)]e^{-\frac{t}{\tau}}$$

$$i(\infty) = \frac{13.5}{3+1.5}$$

$$i(\infty) = 3 \text{ A}$$

$$\tau = \frac{L}{R_{eq}}$$

$$\tau = \frac{2}{3+1.5}$$

$$\therefore \tau = \frac{2}{4.5} \text{ s}$$

$$i(t) = 3 - [3 - 2]e^{-\frac{4.5t}{2}}$$

$$\therefore \boxed{i(t) = 3 - e^{-\frac{9}{4}t} \text{ A}} \text{ for } t > 0$$